

UDITA RAHMAN ARSHEE

Electronics and Communication Engineering Graduate | Wireless Networks | WSN/IoT | AI for Communications

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RESEARCH INTEREST

Wireless Sensor Networks • Internet of Things • Wireless Communications and Networks • Intelligent Wireless Systems • Network Optimization & Resource Allocation • Energy-Efficient Communication Systems

EDUCATION

B.Sc. in Electronics and Communication Engineering, 2026

Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh

- **CGPA:** 3.26/4.00 | **Final four term GPAs:** 3.65, 3.74, 3.65, 3.70
- **Selected Coursework:** Wireless Communications, Digital Communication, Telecommunication Engineering, Information Theory, Computer Networks, Optical Fiber Communications, Satellite Communications & RADAR, Microwave Engineering, Antenna Engineering

PUBLICATIONS & MANUSCRIPTS

PEER-REVIEWED PUBLICATIONS

[1] A. R. Aditta, P. K. Das, **U. R. Arshee**, and A. B. M. Aowlad Hossain, "SenseStick: A Bangla-English Edge-AI Cane for Road Safety and Biomedical Reliability Monitoring," IEEE BECITHCON, 2026.

MANUSCRIPTS SUBMITTED

[2] **U. R. Arshee**, M. Hossen, and S. I. Mumu, "Event-Driven Hierarchical Deep Reinforcement Learning with Spatial Energy Constraints for Wireless Sensor Network Optimization," manuscript submitted to ICEEICT 2027 — 7th International Conference, 2026.

[3] S. I. Mumu, M. Hossen, and **U. R. Arshee**, "THRIFT: Selective Graph-Teacher Knowledge Transfer for Federated Intrusion Detection in Wireless Sensor Networks," manuscript submitted to ICEEICT 2027 — 7th International Conference, 2026.

[4] Most. U. Jannat, **U. R. Arshee**, M. S. Tanvir, D. Aich, A. K. M. B. Haque, X. Xu, and G. Muhammad, "Adaptive PSO-Weighted Grey Wolf Optimized Deep-Q Network Framework for Balancing Energy Consumption and Anomaly Detection Performance in Industrial Wireless Sensor Networks," manuscript submitted to the IEEE Internet of Things Journal, 2026.

[5] A. K. M. B. Haque, B. Bhushan, Most. U. Jannat, **U. R. Arshee**, and P. Bhattacharya, "Deployment of Smart City Concept: Components, Architecture, Applications, Security Issues and Future Directions," manuscript submitted, 2026.

MANUSCRIPT UNDER REVISION

[6] A. R. Aditta, **U. R. Arshee**, S. M. R. Islam, and N. Taaha, "Transformer-Based Modeling of DNA Enhancers: A Controlled Evaluation of DNABERT and Longformer for Identification and Strength Prediction," manuscript under revision for journal resubmission, 2026.

UNDERGRADUATE THESIS

Spatially-Constrained Event-Driven Hierarchical Deep Reinforcement Learning (SCED-HRL) Framework for Energy Optimization in Wireless Sensor Networks

B.Sc. Thesis, Department of Electronics and Communication Engineering, KUET

- Developed a hierarchical DRL framework combining an event-driven PPO strategic layer with DDQN-based tactical agents for energy-aware WSN control.
- Introduced spatial energy budgeting and feasibility constraints to improve network lifetime and balance energy consumption across sensor regions.
- Evaluated the framework using Intel Berkeley sensor data and synthetic traffic, comparing against baseline WSN control strategies on network lifetime, energy fairness, latency, and packet delivery performance.

SELECTED ACADEMIC PROJECTS

Wireless Communication Laboratory — GSM, Zigbee, Bluetooth & RFID

- Conducted hands-on experiments with GSM, Zigbee, Bluetooth, and RFID communication platforms.

Satellite Link Performance Analysis

- Evaluated satellite-link performance under varying frequencies, signal parameters, and propagation distances, focusing on SNR and path-loss effects.

Dual-Band Microstrip Patch Antenna Design and Simulation

- Designed and simulated a microstrip patch antenna targeting dual-band operation at 3 GHz and 7 GHz.

Design and Analysis of Pre-emphasis and De-emphasis Filters

- Designed and analyzed pre-emphasis and de-emphasis filter characteristics using MATLAB/Simulink for telecommunication signal processing.

RFID-Based Smart Car Parking System

- Developed an IoT-based parking prototype using RFID, ESP8266, Arduino, and Blynk for vehicle identification and monitoring.

TECHNICAL SKILLS

- **Programming & Scientific Computing:** Python, C, SQL, MATLAB, NumPy, Pandas, Matplotlib

- **Machine Learning:** PyTorch, TensorFlow/Keras, scikit-learn, XGBoost; CNNs, GNNs, tree-based models; regression, classification, model evaluation
- **Embedded & IoT Systems:** Raspberry Pi, ESP32, ESP8266, Arduino, RFID, GSM, Zigbee, Bluetooth, OpenCV
- **Computer Networking:** Routing & switching, VLAN configuration, OSPF, ACLs, DHCP, DNS, network design and troubleshooting, Cisco Packet Tracer
- **Engineering Tools:** Simulink, Multisim, Arduino IDE, Visual Studio Code

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Department Executive, KUET Research Society | 2025–Present

- Coordinated research-focused events and supported student participation in technical and academic activities.

Campus Ambassador, SheSTEM | 2025–Present

- Promoted STEM awareness and encouraged student engagement in technical initiatives and events.

CERTIFICATIONS & ACHIEVEMENTS

- **Champion, Technical Writing — EUREKA S/1 — KUET Research Society, 2025**
- Supervised Machine Learning: Regression and Classification — DeepLearning.AI, Coursera
- Computer Networking — EDGE Project, ICT Division, Government of Bangladesh
- Data Analysis & Programming — DataCamp
- Industrial Visits: UGC National Data Center; Bangladesh Cable Shilpa Limited (BCSL); Walton Hi-Tech Industries PLC

REFERENCES

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